

# Bisection method

- **Start**
- Define  $f(x)$
- Choose initial guesses  $x_l$  and  $x_u$
- Set tolerable error  $\epsilon_{\text{pes}}$
- Calculate midpoint  $x_m = x_l + x_m / 2$
- Check  $f(x_m) = 0$ 
  - If  $f(x_2) = 0$ , stop.
  - If  $f(x_l) \times f(x_m) < 0$ , set  $x_u = x_m$ .
  - If  $f(x_0) \times f(x_2) > 0$ , set  $x_l = x_m$ .
- If  $|f(x_m)| \leq \epsilon_{\text{pes}}$ , stop.
- If not, repeat from step 5.
- Output the root approximation  $x_m$

We use the bisection method to find the root of the polynomial:  $f(x) = x^3 - x - 2$

**Step 1: Initial Interval Selection** Choose  $a = 1$  and  $b = 2$ .

Evaluate the function:  $f(1) = -2$ ,  $f(2) = 4$

Since  $f(1)$  and  $f(2)$  have opposite signs, there is a root between 1 and 2.

**Step 2: First Iteration**, Calculate the midpoint:  $c_1 = \frac{1+2}{2}$

Evaluate the function at  $c_1$  :  $f(1.5) = -0.125$

Since  $f(1.5)$  is negative, update the interval to  $[1.5, 2]$ .

**Step 3: Repeat:** Repeat the process until the interval becomes sufficiently small, converging on the root.

